Milestone 02

Answer Key

Environment

Update the YAML with your first and last name.

Load the packages.

```
library(tidyverse)
library(haven)
library(gssr)
library(gssrdoc)
library(summarytools)
```

Load the 2024 GSS data

```
# ADD YOUR CODE HERE TO LOAD THE GSS 2024 DATA
gss24 <- gss_get_yr(2024)
```

You'll be working with the following GSS variables:

- `hrs1`: Hours worked per week
- `tvhours`: Hours spent watching TV per day
- `sex`: Respondent's gender
- `polviews`: Political views (from extremely liberal to extremely conservative)

Code, Output, Meaning

Use R to complete the checkpoints below. Show your work (e.g., R code chunks) where appropriate. Add narrative (text outside code chunks) or comments (text inside the code chunks) throughout.

Reference specific statistics (where appropriate) from your output to justify your answers. Explain what the values tell you about the data; interpret their meaning in relation to the question.

Checkpoint 01: Clean variables

Use `zap_missing()`, `as_factor()`, and `droplevels()` to make `sex` into a clean factor variable.

```
gss24$sex <- zap_missing(gss24$sex)
gss24$sex <- as_factor(gss24$sex)
gss24$sex <- droplevels(gss24$sex)
```

Checkpoint 02: Summarize weekly work hours (`hrs1`)

Use `drop_na` and `sumamrise()` to create a table showing the frequency, median, mean, and sd of **`hrs1`**

BONUS: use `round()` to indicate keeping 2 decimal places.

```
gss24 |>
  drop_na(hrs1) |>
  dplyr::summarize(
    freq = n(),
    median_hrs1 = round(median(hrs1), digits = 2),
    mean_hrs1 = round(mean(hrs1), digits = 2),
    sd_hrs1 = round(sd(hrs1), digits = 2)
    )
```

```
# A tibble: 1 x 4
   freq median_hrs1 mean_hrs1 sd_hrs1
  <int>       <dbl>     <dbl>   <dbl>
1  2201          40      39.3    13.6
```

Is the median value for `hrs1` consistent with the mean? Speculate what might explain any differences.

The median (40) and mean (39.29) are close, suggesting little skew. The slightly lower mean than median suggests a mild left (e.g., negative) skew, indicating some respondents report very low hours (likely due to unemployment or few part-time hours).

Interpret the mean and standard deviation for `hrs1`.

On average, Americans work slightly less than a standard full-time schedule of 40 hours per week (39 hours). The standard deviation of 14 hours indicates 68% of respondents report between 26 and and 53 weekly work hours. This is a substantial amount of variation, as some people work significantly fewer or more hours per week. This likely reflects a mix of part-time, full-time, and overtime work patterns.

Checkpoint 03: Summarize daily tv hours (`tvhours`)

Use `drop_na()` and `summarize()` to create a table showing the frequency, median, mean, and sd of **`tvhours`**.

BONUS: use `round()` to indicate keeping 2 decimal places.

```
gss24 |>
  drop_na(tvhours) |>
  summarize(
    freq = n(),
    median_tv = round(median(tvhours), digits = 2),
    mean_tv = round(mean(tvhours), digits = 2),
    sd_tv = round(sd(tvhours), digits = 2)
    )
```

```
# A tibble: 1 x 4
   freq median_tv mean_tv sd_tv
  <int>     <dbl>   <dbl> <dbl>
1  2590       2.5    3.36  3.37
```

Is the median value for `tvhours` consistent with the mean? Speculate what might explain any differences.

The median shows that the typical American watches 2.5 hours of TV per day. The mean (3.36 hours per day) is higher than the median, which suggests the data is right-skewed (e.g., positively skewed), indicating some people watch much more TV than typical Americans, pulling the average up.

Interpret the mean and standard deviation for `tvhours`.

Americans watch an average of 3.36 hours of TV per day. The standard deviation of 3.37 daily hours suggests a great amount of variability in TV habits. Based on the 68-95-99 rule, about 95% of individuals likely watch TV between 0 and nearly 10 hours daily, showing that some watch none or very little television per day and others spend most of their day watching TV.

Checkpoint 04: Summarize daily tv hours (`tvhours`) by `sex`

Use `drop_na`, `group_by()` and `sumamrize()` to create a table showing the mean and sd of `tvhours` by `sex`.

```
gss24 |>
  drop_na(tvhours, sex)  |>
  group_by(sex) |>
  summarize(
    mean = round(mean(tvhours), digits = 2),
    sd = round(sd(tvhours), digits = 2)
    )
```

```
# A tibble: 2 x 3
  sex     mean    sd
  <fct>  <dbl> <dbl>
1 male    3.27  3.39
2 female  3.43  3.37
```

How did the mean and standard deviation of daily tv hours differ between men and women and what do these differences suggest about gender-based variability in tv viewing patterns?

On average, women reported slightly more daily TV hours (3.43) compared with men (3.27). The difference in variability was very similar (~3.38), with the standard deviation differing by less than 1 minute.

Checkpoint 05: Create a new data frame

Create a new data frame called `my_data` that contains only the four variables listed above. (Hint: use `select()`).

Then, use `head()` to preview `my_data`.

```
my_data <- gss24 |>
  select(hrs1, tvhours, sex, polviews)

head(my_data)
```

```
# A tibble: 6 x 4
  hrs1        tvhours           sex    polviews
  <dbl+lbl>   <dbl+lbl>         <fct>  <dbl+lbl>
1    43          15             male       4 [moderate, middle of the road]
2 NA(i) [iap] NA(n) [no answer] male       3 [slightly liberal]
3 NA(i) [iap] NA(i) [iap]       female     1 [extremely liberal]
4    20           2             male   NA(d) [don't know]
5 NA(i) [iap] NA(i) [iap]       female     4 [moderate, middle of the road]
6 NA(i) [iap]     2             male   NA(s) [skipped on web]
```

Checkpoint 06: Remove rows with missing data

Use `drop_na()` to keep only respondents who have non-missing values for all variables in `my_data`.

Then, use `head()` to preview `my_data`.

```
my_data <- my_data |>
  drop_na()

head(my_data)
```

```
# A tibble: 6 x 4
  hrs1      tvhours      sex    polviews
  <dbl+lbl> <dbl+lbl>    <fct>  <dbl+lbl>
1 43        15           male   4 [moderate, middle of the road]
2 80         1           female 2 [liberal]
3 35         4           female 2 [liberal]
4 40         2           female 1 [extremely liberal]
5 60         0 [0 hours] female 4 [moderate, middle of the road]
6 37         2           female 3 [slightly liberal]
```

Checkpoint 07: Create variables

A. Use `mutate()` and `case_when()` to add a new variable to `my_data` called `work_category` that groups respondents as follows:

- "Part-time" if hrs1 is equal to or more than 0 and equal to or less than 34
- "Full-time" if `hrs1` is 35 or more

BONUS: use `table()` to show the counts of your new variable.

```
my_data <- my_data |>
  dplyr::mutate(
    work_category = case_when(
      hrs1 >= 0 & hrs1 <= 34 ~ "Part-time",
      hrs1 >= 35 ~ "Full-time"
  ))

# table(my_data$hrs1, my_data$work_category)
table(my_data$work_category)
```

```
Full-time Part-time
     1093       315
```

B. Use `mutate()` and `case_when()` to add a new variable to `my_data` called `pol3cat` that groups respondents as follows:

- "Liberal" if `polviews` equals "extremely liberal", "liberal", or "slightly liberal"
- "Moderate" if `polviews` equals "moderate, middle of the road"
- "Conservative" if `polviews` equals "extremely conservative", "conservative", "slightly conservative"

BONUS: use `table()` to show the cross-tab of `pol3cat` and `polviews`

```
my_data <- my_data |>
  dplyr::mutate(
    pol3cat = case_when(
      polviews >= 1 & polviews <= 3 ~ "Liberal",
      polviews == 4 ~ "Moderate",
      polviews >= 5 & polviews <= 7 ~ "Conservative"
  ))

table(my_data$polviews, my_data$pol3cat)
```

```
  Conservative Liberal Moderate
1            0      76        0
2            0     203        0
3            0     181        0
4            0       0      525
```

```
  5          160        0         0
  6          202        0         0
  7           61        0         0
```

Checkpoint 08: Summarize tv hours by work category

Use `group_by()` and `sumamrize()` to create a table showing the frequency, mean, median, and sd of **`tvhours`** for each of the two work-hours groups.

BONUS: use `round()` to indicate keeping 2 decimal places.

```
my_data |>
  group_by(work_category) |>
  dplyr::summarize(
    count = n(),
    mean_tvhours = round(mean(tvhours), digits = 2),
    median_tvhours = round(median(tvhours)),
    sd_tvhours = round(sd(tvhours), digits = 2)
    )
```

```
# A tibble: 2 x 5
  work_category count mean_tvhours median_tvhours sd_tvhours
  <chr>         <int>        <dbl>          <dbl>      <dbl>
1 Full-time      1093         2.65              2       2.66
2 Part-time       315         2.97              2       2.96
```

Which work-hours group reported the most daily tv hours in 2024? Was there a lot or a little variability in tv hours?

Part-time workers watched more TV daily (2.97 hours) than full-time workers (2.65 hours), with both groups showing high variability — standard deviations exceeded 2 hours per day.

Is the median value for `tvhours` consistent with the mean? Speculate what might explain any differences.

The mean is greater than the median for both groups, suggesting that there is a positive skew to both distributions. In other words, there are people who watch much more TV than typical of both part-time and full-time workers.

Checkpoint 09: Summarize weekly work and daily tv hours by political identity

A. Use `group_by()` and `sumamrize()` to create a table showing the frequency, mean, median, and sd of **`hrs1`** for each of the three political identity groups.

BONUS: use `round()` to indicate keeping 2 decimal places.

```
my_data |>
  group_by(pol3cat) |>
  dplyr::summarize(
    count = n(),
    mean_hrs1 = round(mean(hrs1), digits = 2),
    median_hrs1 = round(median(hrs1)),
    sd_hrs1 = round(sd(hrs1), digits = 2)
    )
```

```
# A tibble: 3 x 5
  pol3cat      count mean_hrs1 median_hrs1 sd_hrs1
  <chr>        <int>     <dbl>       <dbl>   <dbl>
1 Conservative   423      39.9          40    14.7
2 Liberal        460      38.5          40    13.0
3 Moderate       525      39.2          40    13.4
```

Which political group reported the most work hours in 2024? Was there a lot or a little variability in work hours?

Although Conservatives work more weekly hours (39.87) than Moderates (39.21) or Liberals (38.51) on average, these differences are minor. Additionally, there was great variability in average work hours for each group, with standard deviations exceeding 12 works hours per week. For example, 95% of Conservatives worked between 10 and 69 hours per week (2 SD from the mean).

Is the median value for `hrs1` consistent with the mean? Speculate what might explain any differences.

The median work hours is 40 across all political groups and closely aligns with each group's mean. Liberals show a mean (38.51) slightly lower than the others (39.87 and 39.21), indicating a few more Liberals report working significantly fewer hours than the typical respondent in the other groups.

B. Use `group_by()` and `sumamrize()` to create a table showing the frequency, mean, median, and sd of **`tvhours`** for each of the three political identity groups. **BONUS**: use `round()` to indicate keeping 2 decimal places.

```r
my_data |>
  group_by(pol3cat) |>
  dplyr::summarize(
    count = n(),
    mean_tv = round(mean(tvhours), digits = 2),
    median_tv = round(median(tvhours)),
    sd_tv = round(sd(tvhours), digits = 2)
    )
```

```
# A tibble: 3 x 5
  pol3cat      count mean_tv median_tv sd_tv
  <chr>        <int>   <dbl>     <dbl> <dbl>
1 Conservative   423    2.55         2  2.67
2 Liberal        460    2.66         2  2.68
3 Moderate       525    2.92         2  2.82
```

Which political group watched the most television in 2024? Was there a lot or a little variability in television hours?

Moderates reported the highest average daily TV time (2.92 hours), followed by Liberals (2.66) and Conservatives (2.55). All three groups showed substantial variability, with standard deviations over 2 hours, indicating wide differences in individual viewing habits.

Checkpoint 10: Create a summary dataframe

Use `group_by()` and `summarize()` to create and save a dataframe showing the average work (name it `avg_work_24` and tv hours (name it `avg_tv_24`) for men and women in 2024.

```r
gss24_summary <- my_data |>
  group_by(sex) |>
  dplyr::summarize(
    avg_work_24 = round(mean(hrs1), digits = 2),
    avg_tv_24 = round(mean(tvhours), digits = 2)
    )

head(gss24_summary)
```

```
# A tibble: 2 x 3
  sex    avg_work_24 avg_tv_24
  <fct>        <dbl>     <dbl>
1 male          41.5      2.63
2 female        37.0      2.82
```

Did men or women report more average work hours in 2024? What about TV hours?

Men reported more average work time (42 hours) than women (37 hours), whereas women reported more average TV time (2.82 hours) than men (2.63 hours).

What percentage of men worked 40 hours or less than per week in 2024? (You can either add `R` code chunk to this document or calculate it with the formula. Show your work either way.)

```
my_data|>
  dplyr::mutate(no_overtime = case_when(
    hrs1 <= 40 ~ 1,
    hrs1 >  40 ~ 0,
    TRUE ~ NA_integer_)) |>
  group_by(sex) |>
  summarise(proportion = mean(no_overtime)
)
```

```
# A tibble: 2 x 2
  sex    proportion
  <fct>       <dbl>
1 male        0.662
2 female      0.760
```

About 66% of men worked 40 hours or less than per week in 2024.

IPUMS Data

To keep your Research Brief progress on track, you'll complete short exercises that correspond with the new course material using your own dataset as part of your milestones.

- Present a (pretty) relative frequency table of a nominal or ordinal variable in your dataset.
- Create a summary table for a key interval-ratio (e.g., continuous) variable you are interested in exploring more in your dataset. Include the frequency, mean, median, and sd.

Note: For your final research project, your key dependent variable must be interval-ratio.

Some potentially helpful resources:

- package `ipumsr()`
- Webinar: Using IPUMS data in R with ipumsr
- Just the slides: Using IPUMS data in R with ipumsr